

Retrofitting Springboard Project

April 4th, 2017

Northern Reaches Research – Outline

- Retrofitting springboard project overview
- Northern reaches retrofitting research overview
 - ▣ Identify observed building types in Northern reaches
 - Earthquake damaged
 - Post-earthquake, non-compliant reconstruction
 - ▣ Define retrofit candidate opportunities as initial targets
 - ▣ Pilot



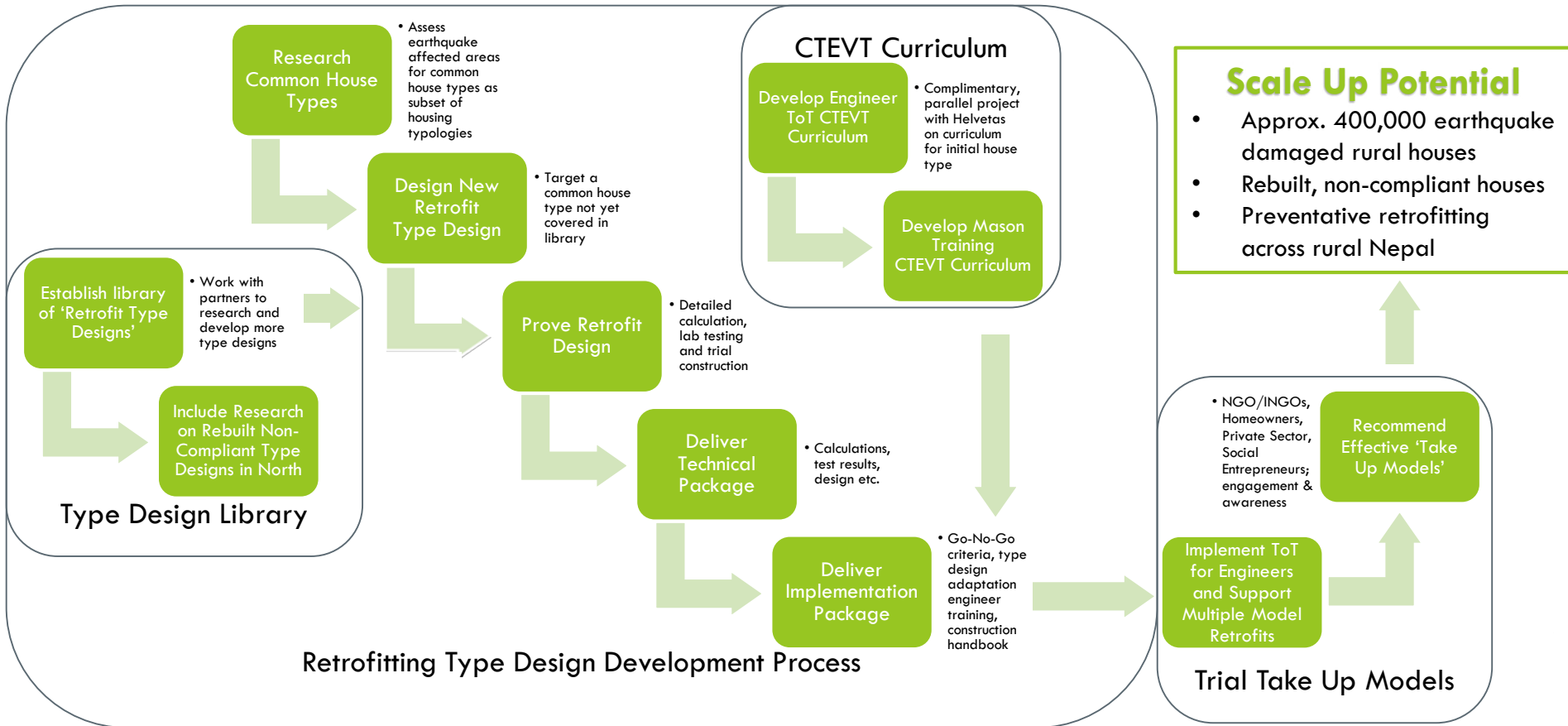
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Retrofitting Springboard Project Overview



Northern Reaches Research – Update

- Initially, building types were observed in Northern reaches of Rasuwa, Gorkha and Dhading
- Main retrofit candidate opportunities identified were;
 - ▣ Traditional dry stone construction (damaged and new)
 - ▣ Wood frame (observations and recommendations to follow)
- Dry Stone Construction
 - ▣ Deficiencies and advantages
 - ▣ Retrofit approaches
- Target prototype building attributes



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Locations Visited

□ Northern Rasuwa

■ Via road

- Goljung
- Gatlang

■ Via trail

- Thuman
- Dalphedi

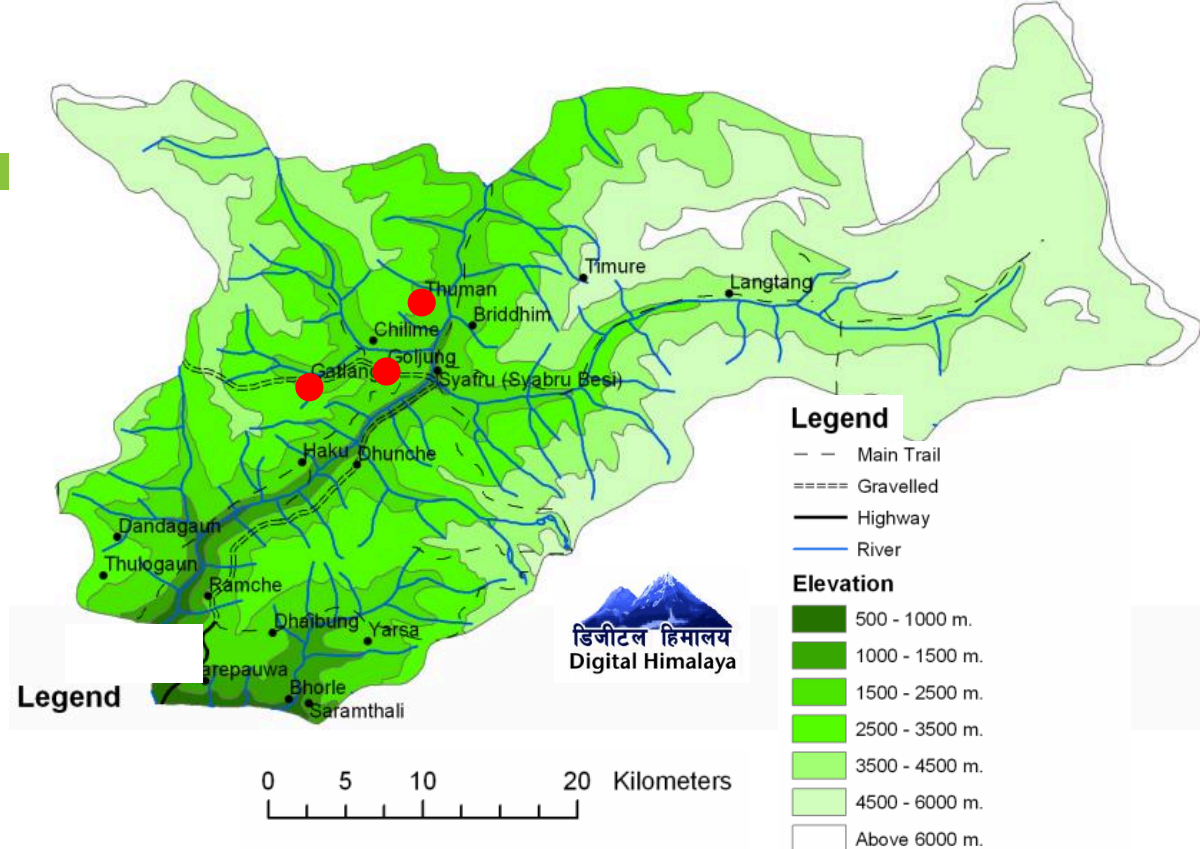
□ Northern Gorkha

□ Via air/ trail

- Samagaon
- Prok
- Nili
- Lokpa
- Philim

□ Northern Dhading

- Lapa



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from the British people

Dry Stone Building Types and Characteristics



Stand alone, most with open fronts.



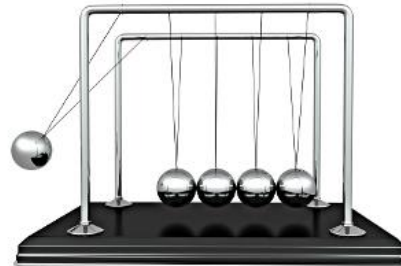
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Dry Stone Building Characteristics



Clustered, adjoining houses

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Dry Stone Building Types and Characteristics



Extensive longitudinal walls and open front.



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Dry Stone Building Characteristics



Heavy roofing



Single storey, open front and heavy roofing.

Dry Stone Building Types and Characteristics



Mud plaster



Steep gables



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Dry Stone Building Characteristics



Unsupported and built in
laminates



Thick dry walls

Dry Stone Building Characteristics



Typical ground floor to
side wall condition



Rafter tie connection



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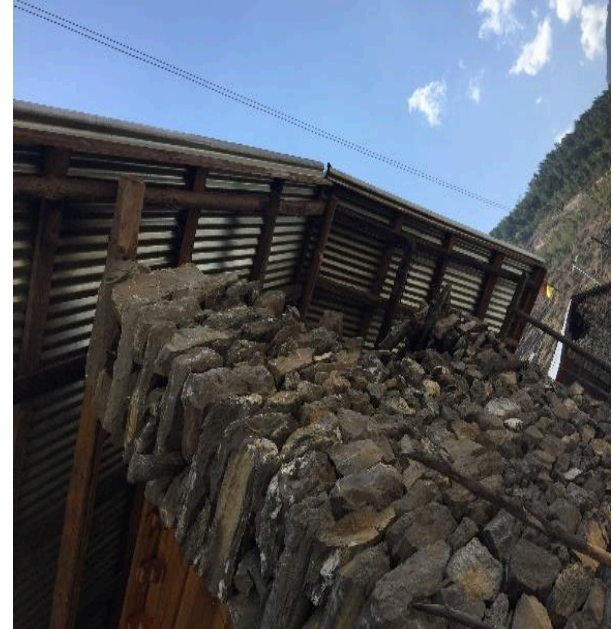


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Traditional Dry Stone Construction-Deficiencies

- Gable End Wall Damage
- Heavy masonry gable walls are unsupported and subject to lateral loads from roof.
- Prone to damage and collapse at low ground motion intensities.



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Traditional Dry Stone Construction-Deficiencies

- ❑ Short Wall Out-Of-Plane
- ❑ Weak attachment between end walls and diaphragms.
- ❑ Damage to and subsequent collapse of end walls due to out-of-plane demands.
- ❑ Wooden slab diaphragm unable to transfer shear loads.



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Traditional Dry Stone Construction-Deficiencies

- ❑ Wall Delamination in both directions
- ❑ The thick walls ($\geq 45\text{cm}$) constructed of interior and exterior facing stones. Required bonding elements are sometimes missing or inadequate.
- ❑ Wall delamination occurs, can lead to subsequent collapse.



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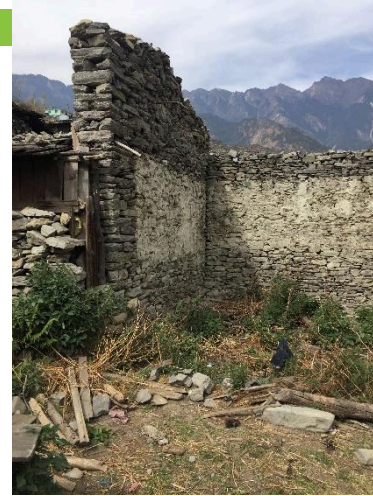


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Traditional Dry Stone Construction – Positive Attributes

- Walls likely to be very strong in-plane
 - Much stronger than mud-and-stone
 - Little evidence of in-plane failures
 - Possibly out-of-plane collapse occurred first
 - Ground floor stone walls at frontage helped reduce complete collapse where present
- Thickness provides rocking resistance out-of-plane
 - Provided wythes tied-together
 - Some well-constructed walls still standing



Traditional Dry Stone Construction – Positive Attributes

- Floor supported independently by wood posts and beams
 - First floor (almost always), Attic (often)
 - Roof (seldom), beams rest on walls
- Large and long timbers
 - Posts 75mm-100mm square or round
 - Beams up to 100mmx150mm
 - Continuous across building
 - No longer available in these lengths
 - Usually undamaged in collapse and reused (robust)
 - Typically in good condition



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Dry Stone Building Retrofit Scheme Recommendations

- Many candidate buildings are close to a road, or have owners of somewhat greater economic means:
 - Propose “heavy” and “light” retrofit scheme, similar to approach for mud-and-stone
 - Difference is extent of reinforced concrete use



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Dry Stone Building Retrofit Scheme Recommendations

- Potentially two “light” schemes, one using more wood and one, less
- Make use of existing wood framing to extent possible
 - Acceptability guidelines can be provided, but not the same assurance as “new”
- Perform testing of critical components to validate retrofit designs
- Note that structural retrofit does not address habitability deficiencies (water, cooking, insulation, etc.)



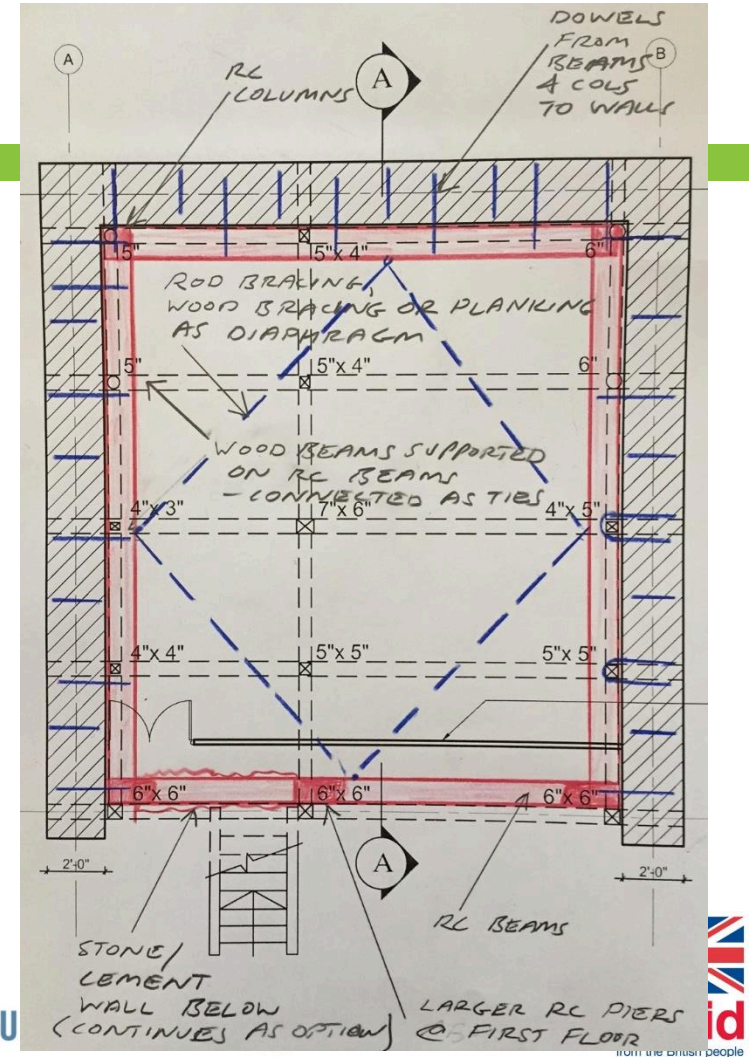
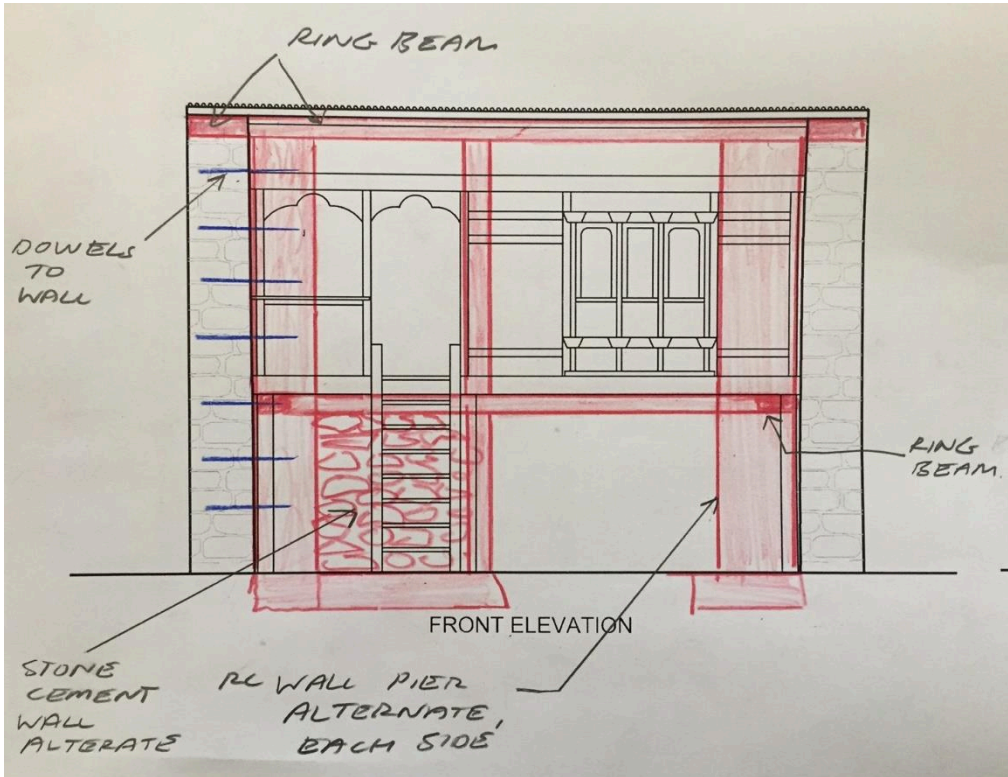
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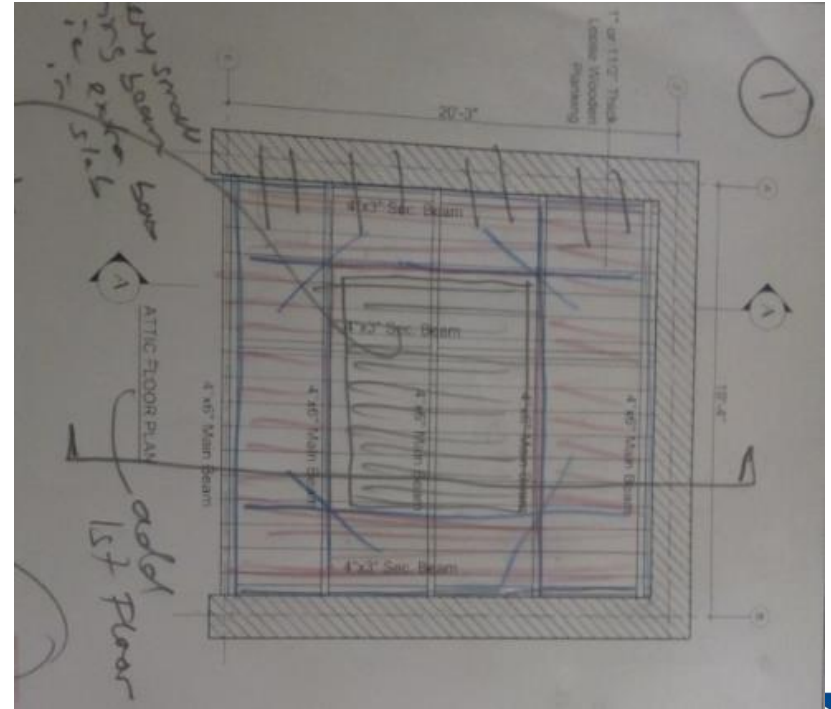
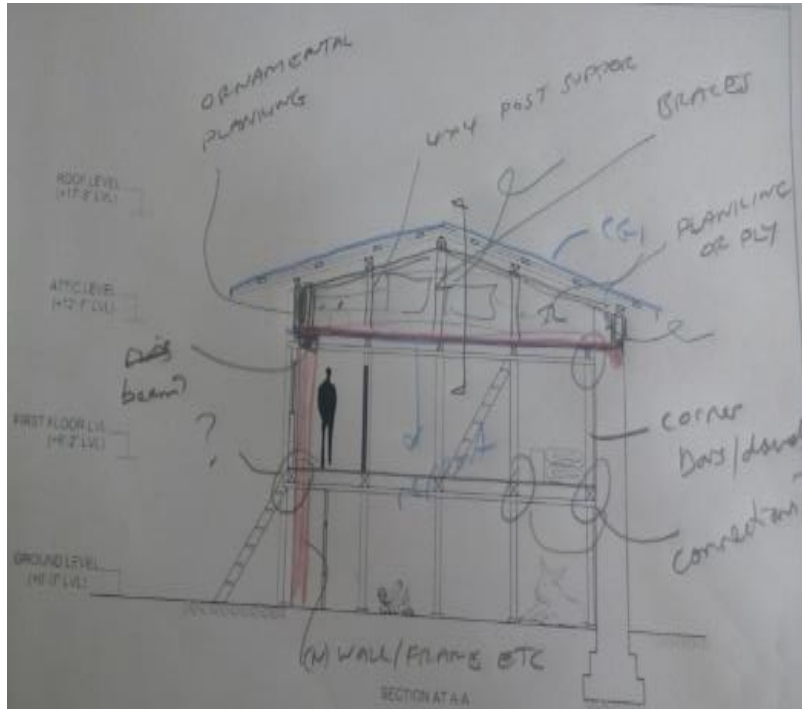
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Heavy Retrofit Draft Concepts



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Target Pilot Building Profile

- “Typical” dry stone construction
 - Size, shape, construction methods and materials
 - Uses the large wood beams and rafters and plank flooring
 - Roofing characters not important (CGI, wood or stone slates)
- Essentially complete
 - Limited earthquake damage
 - No major missing components
- Standalone and clear access
 - Prefer not part of a row, perhaps at the edge of a town
 - Can be built into hillside provided < 1-story, and uphill wall extends to foundation
- Meets usual siting objectives
 - No evidence of foundation failure
 - Not in a flood plane, landslide area, etc., etc.



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